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Exam question

Question

Explain the "sparsity" and "concentration" aspects of the curse of dimensionality, their theoretical justification and their practical impact.

Give examples of the blessing of dimensionality and explain how they are used in practice (you can search for other examples than those presented in the courses or practical work).

Introduction to Dimensional Phenomena

The concept of the **curse of dimensionality** describes the challenges that emerge when working with high-dimensional data. This fundamental notion in data science has two main aspects: **sparsity** and **concentration**, each having strong theoretical justifications and important practical implications.

Interestingly, this same high dimensionality also offers advantages, grouped under the term **blessing of dimensionality**, which can be exploited in various practical applications.

First Part: The Curse of Dimensionality

Data Sparsity

Fundamental concept

Sparsity refers to the extreme dilution of data in high-dimensional spaces. While the volume of the space grows exponentially with the dimension, the number of data points remains limited, creating vast empty regions.

Theoretical justification by exponential volume

Consider a unit hypercube $[0,1]^D$. If we divide each dimension into k equal intervals, we get k^D cells. This exponential growth means that to maintain a constant density of points, we would need an exponential number of samples.

Formally, if we wish to have on average at least one point per cell:

$$N \geq k^D$$

For $D = 100$ and $k = 10$, this would require 10^{100} points, illustrating the practical impossibility of uniformly filling the space.

The empty sphere paradox

A particularly counterintuitive result concerns the ratio between the volume of a hypersphere and that of the hypercube that contains it.

For a hypersphere $B_D(r)$ of radius r centered at the origin, inscribed in a hypercube $C_D(r) = [-r, r]^D$:

Volume of the hypersphere:

$$\text{vol}(B_D(r)) = \frac{2r^D \pi^{D/2}}{D\Gamma(D/2)}$$

Volume of the hypercube:

$$\text{vol}(C_D(r)) = (2r)^D$$

Sphere/cube ratio:

$$\frac{\text{vol}(B_D(r))}{\text{vol}(C_D(r))} = \frac{\pi^{D/2}}{D2^{D-1}\Gamma(D/2)}$$

The key result:

$$\lim_{D\to\infty} \frac{\text{vol}(B_D(r))}{\text{vol}(C_D(r))} = 0$$

This limit shows that in high dimension, almost all the volume of the cube is located in its corners, leaving the central sphere essentially empty.

Practical impact of sparsity

Impossible density estimation: To estimate a probability density in dimension D with a given accuracy, the number of samples required grows exponentially, making non-parametric estimation impractical.

Inefficient k-nearest neighbors algorithm: The "nearest" neighbors are actually very distant, reducing the effectiveness of this method.

Increased risk of overfitting: With few data relative to the number of dimensions, models tend to capture noise rather than the underlying signal.

Problem in information retrieval: Similarity between documents represented by many terms becomes difficult to assess reliably.

Concentration of Measures

Fundamental concept

Concentration refers to the phenomenon where certain quantities, such as distances between points, concentrate around typical values in high dimension, thus losing their discriminatory power.

Theoretical justification by the distance concentration theorem

Let $X \subset \Omega$ be a dataset, a point x and its k nearest neighbors. Denote $D_{\min}$ and $D_{\max}$ the distances to the nearest and farthest neighbors.

Theorem (Beyer et al.): If $\lim_{D\to\infty} \mathrm{Var}\left(\frac{d(x,y)^p}{\mathbb{E}(d(x,y)^p)}\right) = 0$ for $0 < p < \infty$, then for any $\varepsilon > 0$:

$$\lim_{D\to\infty} P\left[\frac{D_{\max} - D_{\min}}{D_{\min}} \leq \varepsilon\right] = 1$$

This formal result shows that in high dimension, all distances between neighbors become relatively similar, making it difficult to distinguish between "close" and "far".

Justification by the Gaussian distribution and the central limit theorem

For independent and identically distributed (i.i.d.) variables $X_i \sim N(0,1)$, consider the Euclidean norm $R = \|X\|_2$. We have:

$$R^2 = \sum_{i=1}^{D} X_i^2 \sim \chi^2(D)$$

Thus:

$$\mathbb{E}[R^2] = D \quad \text{and} \quad \mathrm{Var}(R^2) = 2D$$

The coefficient of variation:

$$\frac{\sqrt{\mathrm{Var}(R^2)}}{\mathbb{E}[R^2]} = \frac{\sqrt{2D}}{D} = \sqrt{\frac{2}{D}} \to 0 \quad \text{when } D \to \infty$$

This convergence, guaranteed by the **Law of Large Numbers**, explains why Gaussian points concentrate on a spherical shell of radius $\sqrt{D}$.

Justification by concentration inequalities

Hoeffding's inequality: For independent random variables $X_1, \dots, X_D$ such that $a \leq X_i \leq b$ almost surely, and for any $\epsilon > 0$:

$$P(|\bar{X} - \mathbb{E}[X]| > \epsilon) \leq 2 \exp\left(-\frac{2D\epsilon^2}{(b-a)^2}\right)$$

where $\bar{X} = \frac{1}{D}\sum_{i=1}^{D} X_i$.

This inequality shows that the empirical mean converges exponentially fast to its expectation, which explains the concentration of measures in high dimension.

Chebyshev's inequality: For a random variable X with expectation μ and variance σ^2:

$$P(|X - \mu| \geq \epsilon) \leq \frac{\sigma^2}{\epsilon^2}$$

When the relative variance decreases with D (as is often the case), the probability of deviation from the mean decreases.

Practical impact of concentration

Loss of spatial discrimination: In clustering algorithms (k-means) and similarity search, the concentration of distances makes it difficult to distinguish between natural groups.

Inefficiency of spatial indexes: Indexing structures like **k-d trees** require exploring a large part of the space, leading to a complexity close to $O(N)$.

Hubness phenomenon: Some points become "hubs", appearing as nearest neighbors to a disproportionate number of other points, thus biasing recommendation systems.

Difficulty in classification: Boundaries between classes become less clear when distances lose their discriminatory power.

Second Part: The Blessing of Dimensionality

Robust convergence of estimators

Law of Large Numbers and Central Limit Theorem

Weak Law of Large Numbers: For i.i.d. variables $X_1, \dots, X_D$ with expectation μ, for any $\varepsilon > 0$:

$$\lim_{D \to \infty} P(|\bar{X} - \mu| < \varepsilon) = 1$$

This law ensures that the empirical mean converges in probability to the theoretical expectation.

Central Limit Theorem (multidimensional version): For i.i.d. random vectors $X_1, \dots, X_D$ in $\mathbb{R}^p$ with expectation μ and covariance matrix Σ, we have:

$$\sqrt{D}(\bar{X}_D - \mu) \underset{d}{\to} N(0, \Sigma) \quad \text{when } D \to \infty$$

This theorem provides not only convergence, but also the limiting distribution, allowing the construction of confidence intervals.

> **Why $\sqrt{D}$?**

1. Setup (clear notations)

Consider random vectors

$$X_1, \dots, X_D \in \mathbb{R}^p$$

They are:

- i.i.d.
- with expectation

$$\mathbb{E}[X_i] = \mu$$

- with covariance matrix

$$\mathrm{Cov}(X_i) = \Sigma$$

Empirical mean

The empirical mean is defined by

$$\bar{X}_D = \frac{1}{D} \sum_{i=1}^{D} X_i$$

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2. Detailed calculation of the expectation

By linearity of expectation,

$$\mathbb{E}[\bar{X}_D] = \frac{1}{D} \sum_{i=1}^{D} \mathbb{E}[X_i] = \frac{1}{D} \cdot D\mu = \mu$$

No surprise.

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3. Detailed calculation of the covariance of $\bar{X}_D$

Definition of matrix covariance

For a random vector $Y \in \mathbb{R}^p$,

$$\mathrm{Cov}(Y) = \mathbb{E}\left[(Y - \mathbb{E}[Y])(Y - \mathbb{E}[Y])^\top\right]$$

Step 1: write $\bar{X}_D - \mu$

$$\bar{X}_D - \mu = \frac{1}{D}\sum_{i=1}^{D}(X_i - \mu)$$

Step 2: develop the covariance

$$\mathrm{Cov}(\bar{X_D}) = \mathbb{E}\left[\left(\frac{1}{D}\sum_{i=1}^{D}(X_i - \mu)\right)\left(\frac{1}{D}\sum_{j=1}^{D}(X_j - \mu)\right)^{\top}\right]$$

We factorize:

$$\mathrm{Cov}(\bar{X_D}) = \frac{1}{D^2}\sum_{i=1}^{D}\sum_{j=1}^{D}\mathbb{E}\left[(X_i - \mu)(X_j - \mu)^{\top}\right]$$

Step 3: use independence

- if $i = j$:
$$\mathbb{E}\left[(X_i - \mu)(X_i - \mu)^{\top}\right] = \Sigma$$
- if $i \neq j$:
$$\mathbb{E}\left[(X_i - \mu)(X_j - \mu)^{\top}\right] = 0$$
(due to independence)

Step 4: count the terms

There are:

- D diagonal terms
- $D(D-1)$ null terms

So:

$$\mathrm{Cov}(\bar{X}_D) = \frac{1}{D^2}\cdot D\Sigma = \frac{1}{D}\Sigma$$

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4. Crucial consequence

$$\boxed{\mathrm{Cov}(\bar{X}_D) = \frac{1}{D}\Sigma}$$

The fluctuations of $\bar{X}_D$ are therefore of size

$$\sqrt{\frac{1}{D}} = \frac{1}{\sqrt{D}}$$

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5. Why we multiply by $\sqrt{D}$

Consider

$$\sqrt{D}(\bar{X}_D - \mu)$$

Then:

$$\mathrm{Cov}!\left(\sqrt{D}(\bar{X}_D - \mu)\right) = D \cdot \mathrm{Cov}(\bar{X}_D) = D \cdot \frac{1}{D}\Sigma = \Sigma$$

Non-degenerate covariance, stable when $D \to \infty$.
Without this factor,

$$\bar{X}_D - \mu \to 0$$

trivially.

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6. Why not D, nor $D^{1/3}$, nor something else

- multiplying by $D \to$ covariance explodes
- multiplying by D^α, $\alpha < \frac{1}{2} \to$ covariance $\to 0$
- multiplying by D^α, $\alpha > \frac{1}{2} \to$ covariance $\to \infty$

$\sqrt{D}$ **is the unique correct scale**.

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7. Alternative formulation of the CLT (expectation & covariance)

General version

Let

$$S_D = \sum_{i=1}^{D} X_i$$

Then:

$$\mathbb{E}[S_D] = D\mu, \qquad \mathrm{Cov}(S_D) = D\Sigma$$

The CLT is written:

$$\boxed{\mathrm{Cov}(S_D)^{-1/2}(S_D - \mathbb{E}[S_D]) \xrightarrow{d} \mathcal{N}(0, I_p)}$$

Version for the mean

Since

$$\bar{X}_D = \frac{1}{D} S_D$$

we obtain:

$$\boxed{\mathrm{Cov}(\bar{X}_D)^{-1/2}(\bar{X}_D - \mathbb{E}[\bar{X}_D]) \xrightarrow{d} \mathcal{N}(0, I_p)}$$

with

$$\mathrm{Cov}(\bar{X}_D) = \frac{1}{D}\Sigma$$

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8. Deep interpretation

The CLT states that:

> After centering by the expectation and renormalization by the square root of the covariance, the sum (or the mean) of independent observations becomes asymptotically Gaussian.

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9. Ultra-clear summary

1. The mean has a variance of $1/D$
2. Fluctuations are of order $1/\sqrt{D}$
3. $\sqrt{D}$ brings the fluctuations to scale
4. The CLT describes their limiting law
5. This scale is **unique**

Final sentence to know how to explain

> The factor $\sqrt{D}$ appears because the covariance of the empirical mean decays like $1/D$. Multiplying by $\sqrt{D}$ is the only renormalization that gives a non-degenerate limit, and leads to a Gaussian law with covariance Σ.

Practical application: Model aggregation

Ensemble learning methods exploit these theorems. For example, **random forests** combine many weakly correlated decision trees. Each tree is an imprecise estimator, but their average benefits from the variance reduction predicted by the central limit theorem.

Example in bioinformatics: With thousands of genes (dimensions) and only hundreds of patients, random forests provide robust classifiers for medical diagnosis.

Random Forests in Medical Genomics - The Art of Intelligent Aggregation

The problem context

In cancer genomics, we often have:

- 20,000 to 60,000 genes (dimensions)
- 100 to 500 patient samples
- A major imbalance: $D \gg N$

The paradox: How can we learn something reliable with so few samples compared to the number of dimensions?

The solution by random forests

First step: Creation of diversity

Each decision tree is trained on:

- A bootstrap sample (67% of patients, randomly chosen)
- A random subset of genes (typically $\sqrt{D} \approx 150$ genes)

Second step: Aggregation by wisdom of crowds

For a new patient:

1. Each tree examines its assigned 150 genes
2. Each tree votes "cancer" or "non-cancer"
3. The final decision is the majority vote

The deep dimensional intuition

Why it works so well:

1. **Local dimension reduction:** Each tree only sees 150 genes out of 20,000 - it operates in a reduced-dimensional subspace
2. **Partial independence:** Trees are decorrelated because they see different genes
3. **Law of large numbers:** With 500 trees, the average error converges to zero

The analogy: Imagine 500 doctors examining a patient, each with a different specialty (cardiology, neurology, etc.). No single one can make a perfect diagnosis, but their consensus is extremely reliable.
Concrete result: In cancer diagnosis machine learning competitions, random forests often beat more complex methods like deep neural networks, because they better exploit data sparsity.

Fraud Detection by Isolation Forest - The Art of Isolation

The problem of fraudulent transactions

In banking transactions:

- 99.9% of transactions are legitimate
- 0.1% are fraudulent
- Each transaction has 100+ characteristics (amount, location, time, merchant type...)

The challenge: Find the needles in the haystack.

The principle of Isolation Forest

Counterintuitive idea: Instead of modeling what is normal, model what is easy to isolate.
In high dimension: Anomalies are easier to "isolate" because they are different from the mass on several dimensions simultaneously.

Algorithm step by step

1. **Subsampling:** Take 256 transactions at random
2. **Tree construction:**
    - Choose a random dimension

- Choose a random split value between min and max
- Separate the transactions
- Repeat until each transaction is isolated

3. **Anomaly score:** The faster a transaction is isolated (few splits needed), the more abnormal it is

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Dimensional intuition

Imagine a crowd of people in a park (normal transactions). Now someone starts shouting and waving their arms (anomaly).
In low dimension: Difficult to spot in the crowd **In high dimension:** The person is different on several axes simultaneously (position, volume, movement…), so easy to isolate
Why it works in high dimension:

- A normal transaction is "average" on all dimensions
- A fraud is "extreme" on at least one dimension
- In high dimension, the probability that a transaction is extreme on several dimensions is very low… except for frauds

Result: Fraud detection with 95% accuracy, while manually examining less than 0.1% of transactions.

Random projections and distance preservation

Johnson-Lindenstrauss lemma

Statement of the lemma: For any set X of N points in $\mathbb{R}^D$ and any $0 < \varepsilon < 1$, there exists a linear map $f : \mathbb{R}^D \to \mathbb{R}^d$ with $d = O(\varepsilon^{-2} \log N)$ such that for all $x, y \in X$:

$$(1-\varepsilon)\|x-y\|^2 \leq \|f(x)-f(y)\|^2 \leq (1+\varepsilon)\|x-y\|^2$$

The dimension d is **independent** of D and depends only logarithmically on N.

Intuitive proof: The proof uses random matrices whose entries are i.i.d. according to a normal distribution, and exploits the concentration of measures (via Hoeffding's inequality) to show that distances are preserved with high probability.

Practical application: Efficient dimension reduction

Approximate nearest neighbor search: Locality-Sensitive Hashing (LSH) algorithms use random projections to quickly index billions of points in high dimension.

Data compression: Compressed sensing techniques allow reconstructing signals from few measurements by exploiting the natural sparsity of real signals.

Concrete example: YouTube's recommendation systems use variants of LSH to quickly find similar videos among billions of possibilities.

YouTube Recommendation and Locality-Sensitive Hashing - The Magic of Random Projections

The technical challenge

YouTube must find similar videos among:

- 800 million videos
- Each video represented by 1000 to 5000 characteristics (metadata, content embeddings)
- Real-time queries (less than 100ms)

The search problem: Find the 10 most similar videos among 800 million in 100ms.

The solution by LSH and random projections

Theoretical foundation: Johnson-Lindenstrauss Lemma
For any set of N points in dimension D, they can be projected into dimension $d = O(\log N)$ while preserving distances.
Practical implementation:

1. **Random projection:** Each video (vector in $\mathbb{R}^{1000}$) is multiplied by a random matrix $R \in \mathbb{R}^{1000\times 50}$
2. **Quantization:** The projected vector is converted to a binary code (hash)
3. **Indexing:** Videos with similar codes are stored in the same "buckets"

Geometric intuition

Why do random projections preserve similarities?
Consider two videos A and B very similar (small distance). After random projection:

- Their projected vectors remain close (with high probability)

- Because random projection approximately preserves distances

The analogy: Imagine projecting 3D objects onto a wall with a flashlight. If two objects are close in 3D space, their shadows will be close on the wall, regardless of the light direction.
Performance gain:

- Without LSH: Compare with 800 million videos → 800 million similarity calculations
- With LSH: Compare with ~1000 videos in the same bucket → 1000 calculations
- **Acceleration: 800,000x**

Quasi-orthogonality and linear separability

Quasi-orthogonality phenomenon

In high dimension, random vectors tend to be almost orthogonal. For two random unit vectors u, v on the sphere in dimension D, the distribution of their dot product concentrates around 0.

Justification by the law of large numbers: If the components are i.i.d. with zero mean, then $\langle u, v \rangle = \frac{1}{D} \sum_{i=1}^{D} u_i v_i$ converges to 0.

Practical application: Support Vector Machines (SVM)

The **kernel trick** allows implicitly projecting data into a very high-dimensional space where they become linearly separable, without explicit calculation of the transformation.

Example in computer vision: For object recognition, SVMs with RBF kernel separate non-linearly separable classes in the original space.

Practical application: Word embeddings

Models like **Word2Vec** project words into spaces of dimension 100 to 300 where semantic relations become linearly representable. For example:

$$\text{"king"} - \text{"man"} + \text{"woman"} \approx \text{"queen"}$$

This property emerges naturally in high dimension thanks to the quasi-orthogonality of different word vectors.

Word Embeddings and the Magic of Semantic Arithmetic

The counterintuitive phenomenon

In word vector spaces (Word2Vec, GloVe), we observe that:

$$\text{"king"} - \text{"man"} + \text{"woman"} \approx \text{"queen"}$$

How is this possible in a 300-dimensional space?

Explanation by high-dimensional geometry

Construction of embeddings:

1. Each word is initialized as a random vector in $\mathbb{R}^{300}$
2. Training adjusts the vectors so that words appearing in similar contexts have similar vectors
3. After training, semantic relations emerge as vector relations

Why it works in high dimension:

1. **Quasi-orthogonality:** Vectors of different words are almost orthogonal
2. **Sufficient space:** 300 dimensions offer enough “room” to encode thousands of relations
3. **Linearity of relations:** Semantic transformations become approximately linear

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Concrete geometric intuition

Imagine that each dimension represents an abstract “concept”:

- Dimension 1: “royalty”
- Dimension 2: “male gender”

- Dimension 3: "female gender"
- Dimension 4: "authority"
- ...

Then:

- "king" = [1.0, 0.9, 0.1, 0.8, ...]
- "man" = [0.1, 0.9, 0.1, 0.3, ...]
- "woman" = [0.1, 0.1, 0.9, 0.3, ...]

The calculation: [1.0, 0.9, 0.1, 0.8] - [0.1, 0.9, 0.1, 0.3] + [0.1, 0.1, 0.9, 0.3] = [1.0, 0.1, 0.9, 0.8]
Which corresponds to "queen" = [1.0, 0.1, 0.9, 0.8, ...]
Real application: Google Translate uses this property to translate between languages without an explicit dictionary, by aligning the vector spaces of different languages.

Statistical robustness and implicit regularization

Double descent phenomenon

Contrary to classical intuition, increasing a model's capacity (number of parameters) beyond a certain point can improve generalization, a phenomenon observed in deep neural networks.

Theoretical explanation: In very high dimension, models benefit from implicit regularization due to the concentration of measures and the particular geometry of high-dimensional spaces.

Practical application: Deep learning

Neural networks with millions of parameters often generalize better than smaller models, especially when combined with appropriate regularization techniques.

Example: Large-scale language models like GPT-3, with 175 billion parameters, achieve remarkable performance in text understanding and generation.

Synthesis and Perspectives

Dialectic between curse and blessing

These two aspects are not contradictory but complementary. The key lies in understanding the conditions under which each phenomenon dominates:

- The **curse** predominates when data is uniformly distributed and dimensions are independent
- The **blessing** emerges when data possesses an underlying structure (sparsity, low-dimensional manifolds) that can be exploited

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Implications for data science practice

Strategies to mitigate the curse

- **Dimension reduction** (PCA, t-SNE, UMAP)
- **Regularization** (ridge, lasso) to avoid overfitting
- **Feature selection** to keep only relevant dimensions
- **Specific algorithms** designed for high dimension (isolation forests, one-class SVM)

Strategies to exploit the blessing

- **Random projections** for indexing and compression
- **Kernel methods** for linear separability
- **Model aggregation** for statistical robustness
- **Representation learning** to discover underlying structure

Unifying principles

The resolved paradox

The "blessing" emerges when we stop fighting high dimensionality and start exploiting it:

1. **Sparsity becomes a strength:** Compressed Sensing
2. **Concentration becomes a guarantee:** Random projections
3. **Law of large numbers becomes an accelerator:** Random forests
4. **Quasi-orthogonality becomes an organization:** Word embeddings

The paradigm shift

Instead of: "How to reduce dimension to avoid problems?"

We adopt: "How to exploit high dimension to solve problems?"

The essence of the blessing

In one sentence: High dimension gives us enough space so that hidden structures become linear, separable, and exploitable by simple but massively parallel algorithms.

This understanding transforms the curse into an opportunity, paving the way for elegant solutions to some of the most complex problems in modern data science.

Recap of theoretical foundations

For the curse:

- Exponential volume growth (sparsity)
- Distance concentration theorem (Beyer et al.)
- Concentration of Gaussian norms (Law of Large Numbers)
- Concentration inequalities (Hoeffding, Chebyshev)

For the blessing:

- Law of Large Numbers and Central Limit Theorem
- Johnson-Lindenstrauss lemma (distance preservation)
- Quasi-orthogonality of random vectors
- Linear separability in high dimension

Conclusion

High dimensionality presents a complex landscape where formidable challenges and precious opportunities coexist. **Sparsity** and **concentration**, although problematic for many classical methods, can become allies when understood and judiciously exploited.

The modern data scientist must navigate this duality by mobilizing both a deep theoretical understanding (concentration theorems, laws of large numbers, geometric properties) and an arsenal of practical techniques (dimension reduction, regularization, adapted methods).

The essential lesson is that dimensionality is not a fate but a characteristic of the problem whose implications must be understood to transform obstacles into opportunities. This dual understanding is crucial for designing systems that thrive in the high-dimensional spaces that increasingly characterize contemporary data.